

Bipeng Wang

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SUMMARY

Ph.D. in Chemical Engineering with interdisciplinary expertise in artificial intelligence, computational materials science, and scientific computing. Developed machine learning methods and AI-driven research tools for atomistic simulations, multimodal learning, and scientific data analysis. Experienced in integrating physics-based modeling, machine learning, and large-scale computing to accelerate engineering and materials research.

EDUCATION

09/2019-08/2025 **Ph.D. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*

01/2021-12/2023 **M.S. (Computer Science)**, *Univ. of Southern California, Los Angeles, USA*

09/2017-05/2019 **M.S. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*

09/2013-05/2017 **B.S. (Applied Chemistry)**, *Shenyang Univ. of Chemical Technology, Shenyang, China*

PROJECT EXPERIENCE 2019-present

AI for Scientific Discovery and Engineering

- Developed AI-powered scientific assistants integrating large language models (LLMs), retrieval-augmented generation (RAG), vector databases, and domain-specific scientific workflows for automated knowledge discovery and data interpretation.
- Designed end-to-end AI pipelines for scientific data processing, semantic retrieval, visualization, and automated report generation, significantly reducing manual analysis effort.
- Built scalable backend services and interactive web applications using FastAPI, enabling researchers to perform AI-assisted analysis through intuitive interfaces.
- Integrated machine learning, information retrieval, and scientific computing into unified AI workflows that support data-driven decision-making across engineering and materials research.

Machine Learning for Materials and Physical Systems

- Developed machine learning surrogate models for atomistic simulations, achieving over 100× acceleration while maintaining high predictive accuracy for scientific applications.
- Designed deep learning architectures for long-horizon time-series prediction, sparse signal reconstruction, and representation learning from molecular simulation trajectories.
- Applied uncertainty quantification, anomaly detection, and sequence modeling techniques to improve the robustness, interpretability, and efficiency of scientific machine learning models.
- Built scalable machine learning pipelines using PyTorch for scientific data analysis, enabling efficient modeling of complex materials and engineering systems.

Computational Materials Modeling

- Investigated charge-carrier dynamics, defect physics, and excited-state processes in functional materials using first-principles simulations and atomistic modeling.
- Combined density functional theory (DFT), molecular dynamics (MD), and machine learning to understand electronic structure evolution and materials properties across multiple time scales.
- Developed computational workflows integrating electronic structure calculations, atomistic simulations, and AI models to accelerate materials discovery and engineering research.

SKILLS

Artificial Intelligence: Machine Learning, Deep Learning, Scientific Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multimodal Learning, Representation Learning, Time-Series Modeling, Uncertainty Quantification

Scientific Computing: Computational Materials Science, Density Functional Theory (DFT), Electronic Structure Calculations, Molecular Dynamics (MD), Nonadiabatic Molecular Dynamics (NAMD), High-Performance Computing (HPC)

Programming & Data Science: Python, PyTorch, NumPy, SciPy, Pandas, Scikit-learn, FastAPI, SQL, Linux, Git

Scientific Software & Simulation: VASP, CP2K, LAMMPS, DeepMD-kit, ASE, pymatgen, Libra, PyXAID

PUBLICATIONS

Machine Learning Model Development:

- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Artificial Neural Networks**. *J. Phys. Chem. Lett.* **2021**, *12*, 6070-6077. ([Link](#))
- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Bidirectional Long Short-Term Memory Networks**. *J. Phys. Chem. Lett.* **2023**, *14*, 7092-7099. ([Link](#))
- **Wang, B. P.** et al. Identifying Rare Events in Quantum Molecular Dynamics of Nanomaterials with Outlier Detection Indices. *J. Phys. Chem. Lett.* **2024**, *15*, 10384-10391. ([Link](#))

Scalable Computational Frameworks:

- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Inverse Fast Fourier Transform**. *J. Phys. Chem. Lett.* **2022**, *13*, 331-338. ([Link](#))
- **Wang, B. P.** et al. Efficient Modeling of Quantum Dynamics of Charge Carriers in Materials Using Short Nonequilibrium Molecular Dynamics. *J. Phys. Chem. Lett.* **2023**, *14*, 8289-8295. ([Link](#))
- **Wang, B. P.** et al. Electron-Volt Fluctuation of Defect Levels in Metal Halide Perovskites on a 100 Ps Time Scale. *J. Phys. Chem. Lett.* **2022**, *13*, 5946-5952. ([Link](#))
- **Wang, B. P.** et al. Sub-Bandgap Charge Harvesting and Energy up-Conversion in Metal Halide Perovskites: Ab Initio Quantum Dynamics. *Npj Comput. Mater.* **2025** ([Link](#))